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Review

The Use of Dual-Energy X-ray Absorptiometry (DEXA) in Evaluating Recovery After Musculoskeletal Injuries in Athletes

A Scoping Review

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Background: Dual-energy x-ray absorptiometry (DEXA) is a widely used imaging modality in sports medicine and orthopaedics due to its accuracy in assessing bone mineral density (BMD) and body composition. Tracking these parameters provides valuable insights into the recovery process after musculoskeletal injuries, which are prevalent among athletes.

Purpose: To evaluate how DEXA imaging has been used to guide rehabilitation in both recreational and competitive athletes by monitoring changes in BMD and body composition after musculoskeletal injuries.

Study Design: Scoping review; Level of evidence, 4.

Methods: A comprehensive literature search was conducted in Ovid MEDLINE, EMBASE, Cochrane Central, and SportDISCUS databases (2000-2024). Studies were included if they used DEXA to measure BMD or body composition changes in athletes after musculoskeletal injuries. Titles, abstracts, and full texts were screened independently by 2 reviewers, with discrepancies resolved through discussion. Data were synthesized qualitatively, and major trends were reported and identified.

Results: Of 1132 unique records, 12 studies met inclusion criteria, involving 319 athletes (34% female) with injuries such as anterior cruciate ligament (ACL) tears, lumbar stress fractures, femoroacetabular impingement, and Achilles tendon ruptures. Athletes represented a range of sports, including soccer, basketball, cricket, and triathlon, at both competitive and recreational levels. DEXA identified significant BMD declines after injury, particularly in the affected limb. For instance, up to a 7% BMD reduction was observed in surgical limbs after ACL reconstruction, persisting for up to 2 years in some cases. Rehabilitation strategies incorporating blood flow restriction therapy or combined running and isometric exercises better preserved lean mass and BMD compared with conventional programs focused solely on closed and open kinetic chain exercises.

Conclusion: The review demonstrated that DEXA imaging is a promising tool in orthopaedic sports medicine for assessing injury-related changes in BMD and body composition. It provides detailed insights into recovery processes and aids in tailoring rehabilitation strategies and return-to-sport decisions. Some studies incorporated advanced rehabilitation methods such as blood flow restriction therapy, which appeared to accelerate recovery as measured by DEXA outcomes. Future studies should explore integrating DEXA findings into athlete-specific rehabilitation protocols with validated thresholds for return to play.

Keywords: dual-energy x-ray absorptiometry; DEXA; body composition; rehabilitation; return-to-play

Dual-energy x-ray absorptiometry (DEXA) imaging is widely recognized for its accuracy in measuring bone mineral density (BMD) and body composition, providing

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crucial data on skeletal health in various medical contexts. DEXA scans use low-dose x-rays to quantify the mineral content per unit area of bone, offering precise assessment of BMD, bone mineral content (BMC), and bone area. Furthermore, DEXA scans can provide precise measurements in lean and nonlean mass in specific areas of the body, allowing for an understanding of asymmetry. In addition to collecting this scope of data, DEXA is a rapid imaging modality and uses little ionizing radiation (effective radiation dose of 3 μGy per scan).[16] Although traditionally used in the diagnosis and management of osteoporosis, DEXA has increasingly found applications in broader clinical contexts such as sports medicine and orthopaedics, particularly in the evaluation and management of musculoskeletal injuries sustained by athletes.[5,10] Given the high physical demands placed on athletes, maintaining optimal bone and muscle health is critical, especially during the rehabilitation and recovery phases after an injury or surgery.[22] Athletes face unique recovery challenges not encountered in the general population, including the pressure for expedited return-to-play timelines, intense physical loads, and sport-specific biomechanical demands. For example, sports that routinely require asymmetrical performance between left and right sides, such as throwing sports, stick/racquet/club sports, and soccer, may necessitate tailored rehabilitation strategies that target limb-specific recovery. Additionally, training volume, seasonal competition schedules, and psychological performance pressures often require clinicians to use highly sensitive tools like DEXA to monitor microlevel changes in musculoskeletal health that could affect performance or motivation or predispose the athlete to reinjury. The precise, rapid, and noninvasive nature of DEXA makes it particularly attractive for monitoring changes in bone and muscle composition over time, offering potential benefits in individualizing rehabilitation strategies.

Injuries to the upper or lower extremities are common in athletes and can significantly affect performance, in both the short and long term.[31] Effective rehabilitation aims not only to restore function but also to optimize physical conditioning and nutrition to prevent future injuries. DEXA imaging offers a unique advantage in this context by providing detailed insights into changes in BMD, muscle mass, and body composition, which are essential parameters in the recovery process.[18] Early and accurate assessment of changes in bone density and muscle mass during the recovery process could provide critical information to guide rehabilitation, ensuring a return to optimal function and minimizing the risk of reinjury or long-term deficits.[27]

This scoping review examined the existing evidence on the use of DEXA imaging to guide rehabilitation and recovery in athletes who have sustained musculoskeletal injuries. The review focused on how DEXA has been used to assess the effect of these injuries on bone and muscle composition, its role in informing and adjusting rehabilitation protocols, and its effectiveness in improving recovery and return-to-sport outcomes. By synthesizing the current literature, we aimed to clarify the value of DEXA in sports rehabilitation, highlighting its potential benefits and identifying areas for future research.

METHODS

Study Design

The purpose was to summarize available evidence on the use of DEXA in evaluating recovery after sports injuries in athletes, identify gaps in the literature, and make recommendations for future use. Specific outcomes of interest included changes in BMD and body composition (lean mass and nonlean mass) during the recovery process as well as return-to-play and functional/biomechanical outcomes. The PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) extension for scoping reviews (PRISMA-ScR) checklist was followed for the reporting of items in this review.[29]

Search Strategy

A comprehensive search was conducted in the Ovid MEDLINE(R), EMBASE, Cochrane Central Register of Controlled Trials, and SportDISCUS from 2000 to 2024 with no restriction on language of publication. An experienced medical librarian developed the database-specific search strategies (Table 1). The reference lists of all eligible studies and related systematic reviews were also searched for additional eligible studies.

Eligibility Criteria

Primary studies that met the following inclusion criteria were included in this scoping review: (1) The population studied consisted of athletes who sustained musculoskeletal injuries; (2) DEXA measurements were performed at baseline (shortly after injury); (3) at least 1 follow-up DEXA measurement was taken at any time point after

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TABLE 1
Characteristics of Included Studies[a]

Study	Country	Study Design	Population	Type of Injury	Study Group and Description	Patients, N	Female Patients, n (%)	Age, y, Mean ± SD	BMI, kg/m^2, Mean ± SD	Limitations
Knurr et al 2023[13]	United States	Retrospective cohort review	Division 1 collegiate athletes playing American football, soccer, basketball, spirit, track and field, crew, softball, volleyball, wrestling, and tennis	ACL tear	Intervention A: athletes with ACLR tested with isometric knee extensor exercises	43	21 (49)	20.4 ± 1.5	25.7 ± 3.8	Potential selection bias due to inclusion based on available DEXA scans. Additionally, nonstandardized rehabilitation and missing follow-up data introduce variability.
					Intervention B: athletes with ACLR tested while running	54	26 (47)	20.6 ± 1.6	26.6 ± 4.5	
Jack et al 2023[9]	United States	Randomized controlled trial	Competitive and recreational athletes	ACL tear	Intervention: rehabilitation with blood flow restriction therapy	17	5 (30)	28.1 ± 7.4	25.2 ± 2.8	Small sample size from a single institution, lack of long-term follow-up, and unmonitored nutritional intake. Lack of blinding of participants and healthcare providers.
					Control: routine rehabilitation	15	8 (53)	24.1 ± 7.2	26.9 ± 5.3	
Alway et al 2022[1]	United Kingdom	Cross-sectional and cohort	Professional cricket players (fast bowlers)	LSF	Intervention: injured players receiving LSF rehabilitation (details unspecified)	17	0 (0)	19.5 ± 2.33	23.57	Small sample, inconsistent scan timing, and lack of training workload data. Potential confounders such as preexisting BMD loss, diagnosis timing, and individual differences in technique and nutrition.
					Control: uninjured players	12	0 (0)	20.63 ± 2.98	23.94	
Knurr et al 2022[12]	United States	Longitudinal cohort review	NCAA Division 1 athletes	ACL rupture	Intervention: athletes who underwent ACLR	33	13 (39)	21.2 ± 1.2	26.8 ± 4.2	Use of full-body DEXA scans, which lack precision for local BMD estimates. Lack of control over activity levels and missing data at 24 mo. Single-arm study.
Anderson et al 2019[2]	United Kingdom	Case study	Professional soccer player	ACL rupture	Intervention: rehabilitation program with tailored diet and exercises after ACL injury	1	0 (0)	26	24	The single-subject design limits generalizability and control over confounding factors.
Kroker et al. 2019[14]	Canada	Retrospective cohort study	Young adults attending a sports injury clinic	ACL tear	Intervention: rehabilitation and immediate weightbearing	15	10 (66)	31	Women 23, men 26.1	Small sample size due to attrition, reconstructive surgeries, and strict imaging constraints, reducing the ability to model confounders and potentially introducing selection bias. Single-arm study.
Kiebzak 2018[11]	United States	Case report	Recreational tennis player	"Knee injury"	Intervention: diet program, exercise on a treadmill and stationary bike	1	NR	NR	27.5	The single-subject design limits generalizability and control over confounding factors.
Mueller et al 2015[20]	Switzerland	Case study	Professional female ironman triathlete	Femoroacetabular impingement with a labral tear	Intervention: hip arthroscopy, followed by immobilization and gradual increase in training	1	1 (100)	32	NR	The single-subject design limits generalizability and control over confounding factors.
Milsom et al 2014[19]	United Kingdom	Case study	Professional soccer player	ACL rupture and injury of external meniscus	Intervention: 8-wk immobilization and physical therapy, followed by diet-controlled, weightbearing rehabilitation program	1	0 (0)	24	24.3	The single-subject design limits generalizability and control over confounding factors.

(continued)

TABLE 1
(continued)

Study	Country	Study Design	Population	Type of Injury	Study Group and Description	Patients, N	Female Patients, n (%)	Age, y, Mean ± SD	BMI, kg/m^2, Mean ± SD	Limitations
Zerahn et al 2006[32]	Denmark	Case control study	Young adults participating in sports	Chronic ACL insufficiency	Intervention: ACLR and physical therapy	18	5 (28)	32.5 (range, 25-43)	24.3 (range, 18.7-27.7)	Potential confounding factors such as physical activity levels, nutritional status, and baseline bone health were not explicitly controlled.
					Control: Healthy adults	41	18 (44)	30.5 (range, 20-45)	23.3 (range, 17.2-31.1)	
Magnusson et al 2003[17]	Sweden	Prospective cohort study	Competitive athletes	MTSS	Intervention: athletes with MTSS	14	0 (0)	24 ± 3	25.6	Potential confounding factors such as differences in training intensity, nutritional status, and baseline bone health were not fully controlled.
					Control: nonathletes (without MTSS)	24	0 (0)	24 ± 4	24.1	
Therbo et al 2003[28]	Denmark	Prospective cohort study	Recreational athletes	Achilles rupture	Intervention: patients with Achilles rupture undergoing surgical repair and immobilization	12	2 (17)	33	24.1	Small sample size, loss to follow-up for long-term data, and lack of control over confounding factors such as rehabilitation adherence and physical activity. Single-arm study.

[a]ACL, anterior cruciate ligament; ACLR, anterior cruciate ligament reconstruction; DEXA, dual-energy x-ray absorptiometry; LSF, lumbar stress fracture; MTSS, medial tibial stress syndrome; NR, not reported.

the baseline measurement; (4) DEXA measurements were performed near the site of injury. This was defined anatomically as within the specific bone or muscle group directly affected by the injury (eg, the distal femur in the case of ACL injuries, the proximal tibia in tibial stress injuries, or lumbar vertebrae L1-L4 in lumbar spine injuries) to ensure that measurements reflect localized changes in bone and muscle composition relevant to the injury site.

Review papers or studies that did not meet the inclusion criteria above were excluded.

Data Screening and Extraction

Two reviewers (R.D. and U.T.) independently screened titles, abstracts, and full texts for eligibility using Covidence Systematic Review Software (Veritas Health Innovation). Disagreement was resolved by group discussion, with arbitration by a third author (A.H.W.C.) where differences remained. We calculated interrater agreement at the full-text screening stage using the kappa (κ) statistic. The κ value was close to 1, indicating strong agreement between reviewers. The data were extracted and recorded using Microsoft Excel spreadsheets. The following data were extracted from eligible studies: country of publication, study design, population studied, type of injury sustained, intervention details, demographic information (ie, sex, age, body mass index [BMI]), participant sport and level of competition, baseline data (ie, time of baseline testing and data on BMD, body composition, or other notable values), and follow-up testing (ie, BMD, body composition, and other notable outcomes).

Narrative Analysis

This scoping review used a narrative synthesis of the included studies due to heterogeneous study designs, interventions, outcomes, and populations, which precluded meta-analysis. Findings were narratively synthesized and organized by themes related to the research questions, intervention types, and outcome measures, specifically focusing on injury or surgery type, BMD, and body composition trends, and return-to-sport or functional outcomes. Key patterns, similarities, and differences were noted, with attention to study contexts. Factors such as study populations, type of intervention, and outcome assessment methods were explored to explain variations and contradictions in findings.

RESULTS

Of 1132 unique search results, 12 studies that used DEXA to monitor BMD and other body composition changes in a total of 319 athletes (34% female) sustaining various musculoskeletal injuries were included in this scoping review (Figure 1).[#] The following injuries were noted by the included studies: anterior cruciate ligament reconstruction (ACLR) (n = 7 studies),[2,9,12-14,19,32] unspecified "knee injury" (n = 1 study),[11] lumbar stress fracture (n = 1 study),[1] femoroacetabular impingement with labral tear (n = 1 study),[20] medial tibial stress syndrome (MTSS)

[#]References 1, 2, 9, 11-14, 17, 19, 20, 28, 32.

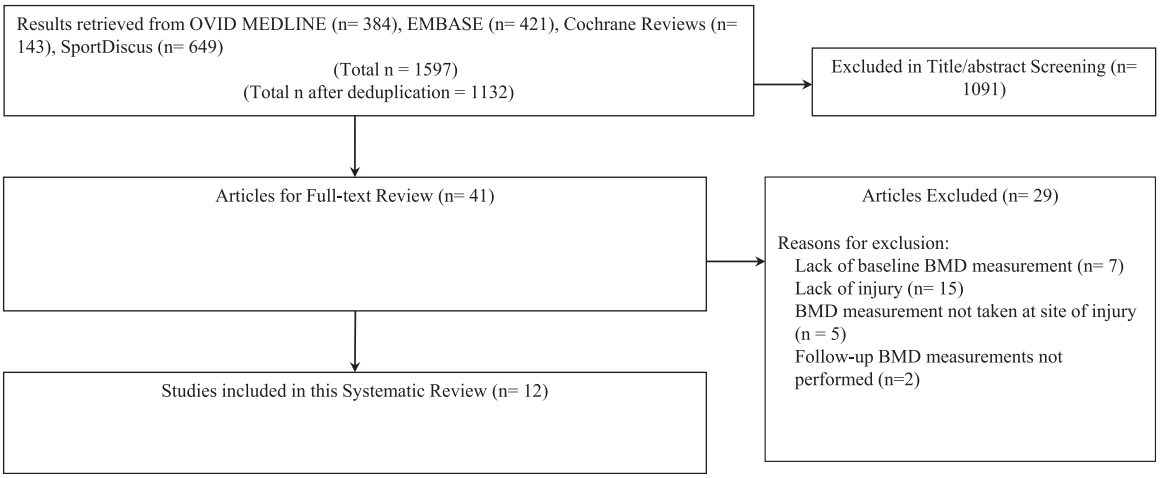

Figure 1. PRISMA flowchart with exclusion details: studies excluded due to lack of injury (n = 15), lack of baseline bone mineral density (BMD) measurement (n = 7), lack of injury site-specific BMD measurement (n = 5), follow-up BMD measurements not performed (n = 2).

(n = 1 study),[17] and Achilles tendon rupture (n = 1 study).[28] The weighted mean age and BMI of athletes across the included studies were 24.5 ± 4.4 years and 25.1 ± 1.2 kg/m^2, respectively. The type of sport played by athletes, as well as level of competition, varied across included studies, with some studies including a diverse range of athletes. The sports played by athletes in the included studies were American football, soccer, basketball, spirit, track and field, crew, softball, volleyball, wrestling, tennis, cricket, and ironman triathlon, as well as unspecified recreational and competitive athletes. A detailed summary of characteristics of included studies can be viewed in Table 1. Furthermore, an overview of outcomes reported by included studies is provided in Table 2.

Bone Mineral Density Changes

Ten of the included studies reported findings on BMD changes using DEXA, examining a range of injuries including ACL tears,[9,12-14,32] lumbar stress fractures,[1] femoroacetabular impingement with labral tear,[19] MTSS,[17] Achilles tendon ruptures,[28] and unspecified knee injuries.[11] In most studies, BMD initially decreases substantially after injury, with varying degrees of recovery over time depending on injury type and intervention. For knee injuries, regions of interest (ROIs) that were found to be reproducible among studies to take BMD measurements were centered at 5% and 15% (F5, F15) of femoral length, measured from the intercondylar notch.[13] In ACL tears,[9,12-14,32] multiple studies showed that BMD decreased in the surgical limb, particularly at the 5% mark of the femur,[12,13] with varying recovery patterns based on rehabilitation type. In general, BMD typically decreased by approximately 4% to 10% within the first 6 months after injury or surgery, with a rapid decline of approximately 0.8% to 1.2% per month during this period.[9,12-14,32] This range represents the mean monthly percentage decline in regional BMD reported across 4 studies examining ACL injuries. Compared with athletes who received traditional rehabilitation with a combination of closed and open kinetic chain exercises, athletes who received blood flow restriction (BFR) therapy showed greater BMD and lean mass measurements by 12 weeks, with lean mass typically recovering faster than BMD.[9] Furthermore, a rehabilitation protocol consisting of running and isometric knee exercises after ACLR led to partial or near-complete recovery in BMD by 25 months.[13] However, a general trend was observed in which BMD deficits were observed around a 12-month follow-up but returned to near baseline levels by 24 months and onward.[12,13,32] In 2 studies, BMD deficits at the 15% femoral level were persistent past 24 months,[12,13] particularly for athletes with lower peak knee extensor movement and RTD.[13] In cricket fast bowlers, lumbar stress fractures exhibited BMD decreases on the contralateral side (the side opposite to the bowling arm) but recovered and surpassed baseline by 52 weeks.[1] A single study examining Achilles tendon ruptures found consistent decreases in BMD across various regions, with a notable 6.8% reduction at the greater trochanter after 1 year.[28] MTSS managed nonoperatively showed a significant increase in BMD in the painful region over 5 years.[17] Although a case of femoroacetabular impingement treated with hip arthroscopy initially resulted in a decrease in BMD during the 2 months of immobilization after surgery, significant recovery was observed within 6 weeks of rehabilitation involving cycling, swimming, and resistance training, with full recovery to presurgery levels achieved within 9 months.[20] Recovery timelines varied significantly, with some injuries like ACL tears showing partial recovery within 12 to 25 months, whereas others, such as MTSS, demonstrated

TABLE 2
Overview of Outcomes Reported by Included Studies

Study	Reported on Bone Mineral Density, Bone Mineral Content, or Bone Mass	Reported on Body Composition	Reported Return-to-Play or Functional Outcomes
Knurr et al 2023[13]	X		X (Isometric knee extensor exercises and running)
Jack et al 2023[9]	X	X	X (Time to return to sport; functional leg exercises)
Alway et al 2022[1]	X		
Knurr et al 2022[12]	X		
Anderson et al 2019[2]		X	
Kroker et al 2019[14]	X		
Kiebzak 2018[11]	X	X	
Mueller et al 2015[20]	X	X	
Milsom et al 2014[19]		X	
Zerahn et al 2006[32]	X		X (Knee performance on sport score)
Magnusson et al 2003[17]	X		
Therbo et al 2003[28]	X		

BMD increases over several years. A detailed summary of BMD outcomes reported by included studies is provided in Appendix Tables A1 and A2.

Body Composition Changes

Five of the included studies used DEXA to report on body composition changes among athletes with injuries. These studies examined athletes after ACLR,[2,9,19] an unspecified "knee injury" managed through dietary and exercise interventions on a treadmill and stationary bike,[11] and femoroacetabular impingement with labral tear treated with hip arthroscopy.[20] Across the 5 studies, body composition changes showed consistent patterns after injury and rehabilitation. Lean mass typically decreased during periods of immobilization or inactivity, with the most significant losses observed in the injured limbs.[2,9,11,19,20] Across studies, injured limbs showed approximately 7% to 14% lean mass reductions during immobilization, with partial to full recovery during rehabilitation, whereas unaffected limbs remained stable or experienced minimal changes (<5%).[2,9,11,19,20] Certain interventions such as BFR therapy[9] and rehabilitation programs incorporating exercise and/or diet interventions led to lean mass recovery,[2,11,19,20] with 1 study reporting full recovery by the end of the follow-up period of 9 months.[20] Some studies showed that dietary interventions that focused on achieving higher daily energy intakes (2500-3000 kcal/d)[2] as well as diets with high protein content (30-40 g per meal) and supplementation with a daily multivitamin, vitamin C, β-hydroxy-β-methylbutyrate, creatine monohydrate, and fish oil concentrate[19] may help prevent muscle atrophy and micronutrient deficiencies during recovery. Additionally, studies showed that interventions with early isometric/isotonic contractions[19] and a gradual progression in weightbearing activities may aid in athletes' recovery.[19,20] Fat mass generally decreased during initial recovery phases but tended to normalize or remain slightly lower after rehabilitation as seen during final follow-up periods between 5 and 12 months.[2,11,19] Overall, interventions that combined structured rehabilitation and dietary adjustments facilitated improvements in lean mass and reductions in fat mass as determined by DEXA. A detailed summary of body composition outcomes reported by included studies is provided in Appendix Tables A1 and A2.

Return-to-Play and Functional Outcomes

Three studies commented on return-to-play and functional outcomes among the injured athletes. In the 2023 study by Knurr et al,[13] athletes with ACLR were either tested with isometric knee extensor exercises (to study peak torque [PT] and rate of torque development [RTD]) or were assessed while running (to study peak knee flexion [PKF] or peak knee extensor movement [PKEM]) so the authors could examine associations between exercise performance and biomechanics with BMD changes. No significant interactions were found between time after ACLR and PT, RTD, PKF, or PKEM for 5% femur BMD. However, significant interactions were observed for these variables with 15% femur BMD, indicating that athletes with lower RTD and PKEM at around 9 to 10 months after ACLR experienced decreases in BMD at the distal femur over time. In the study by Jack et al,[9] time to return to sport was significantly shorter in the BFR group (mean ± SD, 6.42 ± 0.31 months) compared with the routine rehabilitation group (8.30 ± 0.54 months). Functional assessments included single-leg (SL) squat, SL eccentric step-down, SL press, SL hamstring curl, and Y balance. Both groups showed similar improvements in all functional measures over 4 weeks, except for the anterior Y-balance, where only the BFR group had a significant improvement between 8 and 12 weeks. This corresponded to their higher DEXA BMD scores compared with the control group. Zerahn et al[32] found that at 24-month follow-up after ACLR and structured rehabilitation, athletes' knee performance on sport score increased from a median of 2 (abnormal) to 1 (almost normal). Knee performance on activity of daily living increased from a median of 2 (abnormal) to 0 (normal). This improved knee performance in sports activities was

associated with increases in BMD *Z* scores for both the medial and lateral tibia. A detailed summary of return-to-play and functional outcomes reported by included studies is provided in Appendix Tables A1 and A2.

DISCUSSION

This review represents the first effort to systematically consolidate fragmented and heterogeneous evidence on the role of DEXA imaging in sport-specific recovery from musculoskeletal injuries. Although various studies have explored DEXA in isolated contexts, no previous synthesis has integrated findings across injury types, athletic populations, and recovery outcomes. By bringing together this dispersed literature, the present review offers a comprehensive understanding of how DEXA can inform individualized rehabilitation strategies, monitor localized bone and muscle changes, and contribute to return-to-play decision making in athletes.

As DEXA can provide highly accurate, region-specific measurements of BMD and body composition changes, healthcare professionals can use it to monitor injury effects and recovery progress over time.[7] This capability is particularly valuable in sports medicine, where athletes' ability to resume training and competition with minimized reinjury risk is essential. Furthermore, the capability of DEXA to effectively identify sport-specific bone density adaptations and asymmetry in athletes highlights its relevance in sports medicine for monitoring both physiological and pathological changes.[4]

The evidence reviewed consistently shows that BMD often decreased after injury or surgery, followed by time off sport, particularly in the affected limb. For instance, ACL injuries demonstrated a notable reduction in BMD in the injured limb, with partial to near-complete recovery observed over 12 to 25 months after injury.[12,13] Similarly, a case-control study by Bayar et al[3] involving recreational athletes with complete ACL ruptures demonstrated significant region-specific BMD loss in ACL-deficient knees, particularly in the lateral femoral condyles and patella, which was associated with injury duration and activity levels. This aligns with our findings and with other studies on localized BMD reduction,[24] highlighting the potential of targeted rehabilitation to mitigate these deficits. However, studies indicate that certain factors, such as lower peak knee extensor torque, can delay recovery of BMD at specific femoral sites, underscoring the importance of individualized rehabilitation that takes into account an athlete's baseline strength and injury severity.[13] In cases of lumbar stress fractures, BMD recovery exceeded baseline levels by approximately 52 weeks, suggesting that specific injury types may even benefit from targeted rehabilitation that promotes bone density gains beyond preinjury status.[1] These findings infer that DEXA's detailed monitoring of BMD may guide adjustments in rehabilitation intensity and duration to optimize recovery and minimize long-term deficits. Similarly, a prospective longitudinal study involving athletes with tibial bone stress injuries used high-resolution peripheral quantitative computed tomography to measure volumetric BMD, noting an initial decline in volumetric BMD in both injured and uninjured limbs, with recovery to baseline by 24 weeks and exceeding baseline values at 52 weeks, aligning with our observations on lumbar stress fracture recovery.[25]

DEXA has also proven valuable in tracking body composition changes during recovery, especially in assessing lean muscle mass changes in injured limbs. Immobilization after injury typically leads to lean mass losses in the affected limb, whereas targeted rehabilitation, particularly those programs that incorporate resistance training or BFR therapy, can help restore lean mass.[9] The studies included in this review demonstrate that DEXA was effective in quantifying these lean mass changes and monitoring recovery, with some interventions accelerating full recovery of lean mass. However, it should be noted that previous studies have found DEXA to overestimate trunk lean mass compared with magnetic resonance imaging (MRI), particularly in individuals with high intramuscular fat, suggesting the need for careful interpretation of lean mass recovery metrics in specific populations like those with spinal cord injuries.[15,26] As MRI may be impractical in certain settings, DEXA has been validated as a feasible alternative for thigh muscle mass assessment in spinal cord injuries.[15] Moreover, the use of DEXA in evaluating body composition asymmetries may be helpful in understanding injury risk in athletes. A recent cross-sectional study by Vaillancourt et al[30] found significant asymmetries in upper limb and trunk lean body mass among university soccer players, with lower limb asymmetry remaining minimal due to the symmetric demands of running. Although no significant correlations were observed between asymmetries and injury risk, marginal associations were noted between trunk and lower limb asymmetry indices and lower limb injuries, suggesting that body composition asymmetry may be a potential modifiable risk factor for injury prevention in soccer players.[30]

In evaluating functional outcomes and return-to-play timelines, this review suggests that DEXA imaging may contribute indirectly by offering data that informs targeted rehabilitation strategies. For example, Jack et al[9] used DEXA to track lean mass recovery in athletes undergoing BFR therapy after ACLR. Not only did the athletes in BFR group have significantly higher lower extremity lean mass, but they also demonstrated accelerated return-to-play times compared with the non-BFR group, despite similar functional scores at the end of follow-up.[9] In a prospective cohort study of collegiate athletes, lower total body BMD and higher MRI grades (based on periosteal edema on T1- and T2-weighted images) were independent predictors of delayed recovery in bone stress injuries, emphasizing the clinical relevance of integrating BMD assessments into routine conditioning and rehabilitation planning.[21] This underscores the potential role of a comprehensive rehabilitation approach that combines DEXA imaging with functional assessments to ensure both structural and biomechanical readiness for return to sport.

Limitations and Future Directions

Although this review provides valuable insights into the utility of DEXA imaging, several limitations should be noted. The variability in study design, injury types, and rehabilitation protocols limits comparability. Although formal risk of bias assessments were not performed, given the scoping review framework, most included studies were observational in nature with small sample sizes and limited follow-up, which restricts the strength of conclusions drawn. A notable overrepresentation of ACL injury studies skews the evidence base toward lower limb pathologies, whereas upper body injuries—such as shoulder dislocations/instability, rotator cuff tears, and clavicular fractures—remain underrepresented. This imbalance restricts the generalizability of findings across the full spectrum of athletic injuries and underscores the need for further research on upper extremity rehabilitation monitored through DEXA. Additionally, heterogeneous follow-up durations may fail to capture full recovery trajectories or introduce confounding variables. Observational designs prevalent in the literature hinder causal inferences between DEXA-monitored recovery markers (eg, BMD, lean mass) and functional outcomes. Without standardized preseason, postinjury, and rehabilitation DEXA protocols, it remains challenging to establish definitive relationships between DEXA changes and recovery metrics. Inconsistent scanning protocols and preparation methods, such as hydration and positioning variability, may affect measurement reliability. Furthermore, a lack of sport-specific baseline values for BMD and body composition complicates recovery assessments and protocol adjustments.

Hind et al[8] emphasized the need for standardized DEXA scanning protocols and precision error reporting in athletic populations to ensure accurate interpretation of body composition changes during recovery. They highlight that inconsistencies in prescan preparation—such as variability in hydration, fasting, and positioning—can lead to significant errors in lean and fat mass estimates. For example, dehydration may falsely lower lean mass readings, whereas fluid overload could exaggerate them. Improper or inconsistent positioning during the scan can lead to asymmetric data, especially in limb-specific assessments, potentially masking or exaggerating recovery progress. Similarly, differences in limb-specific measurements—if the dominant or injured side is not consistently assessed—can undermine longitudinal comparisons. This reflects the importance of consistent protocols to enhance measurement reliability.[8]

To address these limitations, clinicians may consider adopting provisional thresholds to guide rehabilitation progress. For instance, research indicates that asymmetries >10% in muscle strength or mass are associated with an increased risk of injury and may impede optimal performance. Actionable metrics can improve consistency in monitoring and decision making.

Furthermore, future research should focus on identifying the most appropriate anatomic sites for DEXA assessments tailored to specific injury types, as the optimal site for monitoring BMD may depend on the injury location, rehabilitation goals, and bone characteristics at the affected region.[23] A need remains to establish definitive normative baseline BMD and body composition values for athletes across various sports. Dobrosielski et al[6] emphasized the importance of using DEXA to establish sport-specific baseline and recovery norms, highlighting significant differences in BMD and lean mass between athletes in high-impact sports (eg, gymnastics, basketball) and those in low-impact sports (eg, swimming, cross country). These baselines are crucial for identifying deviations during recovery, allowing for precise adjustments in rehabilitation protocols to ensure athletes return to preinjury levels of BMD and lean mass.

CONCLUSION

Our review demonstrated that DEXA imaging is a promising tool in orthopaedic sports medicine for assessing injury-related changes in BMD and body composition. It provides detailed insights into recovery processes and aids in tailoring rehabilitation strategies and return-to-sport decisions. Some studies incorporated advanced rehabilitation methods, such as BFR therapy, which appeared to accelerate recovery as measured by DEXA outcomes. Future studies should explore integrating DEXA findings into athlete-specific rehabilitation protocols with validated thresholds for return to play. DEXA's ability to detect subtle changes in bone and muscle health may make it a valuable tool for evaluating rehabilitation efficacy and informing clinical decisions in sports medicine.

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APPENDIX

TABLE A1
Summary of Outcomes Examined by Included Studies on ACL Injuries[a]

		Baseline Testing			Follow-up Testing				
Study	Study Group	Time of Baseline Testing After Injury	BMD, g/cm^2, Mean ± SD	Other Notable Baseline Values, Mean ± SD	Follow-up	BMD, g/cm^2, Mean ± SD	Bone Density Change Score (from baseline)	Other Notable Results (eg, return to activity, function)	Summary
Knurr et al 2023[13]	Intervention A (n = 43): athletes with ACLR tested with isometric knee extensor exercises	3 mo	Measured at 5% and 15% marks of the length of femur, measured from distal femur. BMD 5% ROI: surgical limb 1.74 ± 0.27, nonsurgical limb 1.85 ± 0.26 BMD 15% ROI: surgical limb 1.13 ± 0.22, nonsurgical limb 1.19 ± 0.21	PT (N·m·kg^{-1}): surgical limb 2.5 ± 0.7, nonsurgical limb 3.3 ± 0.7 RTD (N·m·kg^{-1}·s^{-1}): surgical limb 7.2 ± 3.3, nonsurgical limb 13.1 ± 3.4	5, 10, 15, 20, 25 mo	Data estimated from graph; SD NR. Surgical limb: 5% femur, **5 mo 1.58, 10 mo 1.6, 15 mo 1.65, 20 mo 1.69, 25 mo 1.71** 15% femur, **5 mo 1.04, 10 mo 1.02, 15 mo 1.01, 20 mo 1.00, 25 mo 0.98** Nonsurgical limb: 5% femur, constant at 1.7 from 5 to 25 mo 15% femur: slight increase from 1.08 to 1.1 over 5-25 mo	NR	No significant interactions were found between time after ACLR and PT, RTD, PKF, or PKEM for 5% femur BMD. However, significant interactions were observed for these variables with 15% femur BMD, indicating that athletes with lower RTD and PKEM at around 9-10 mo after ACLR experienced decreases in BMD over time.	In NCAA Division1 athletes after ACLR, reduced quadriceps performance and altered running mechanics were significantly linked to greater BMD loss in the distal femur. Improving quadriceps strength and running biomechanics after surgery may help prevent this bone loss.
	Intervention B (n = 54): athletes with ACLR tested while running	3 mo	BMD 5% ROI: surgical limb 1.76 ± 0.25, nonsurgical limb 1.85 ± 0.26 BMD 15% ROI: surgical limb 1.13 ± 0.20, nonsurgical limb 1.20 ± 0.21	PKF angle (degrees): surgical limb 37.3 ± 6.0, nonsurgical limb 45.7 ± 4.6 PKEM (N·m·kg^{-1}): surgical limb 1.4 ± 0.5, nonsurgical limb 2.3 ± 0.4	5, 10, 15, 20, 25 mo	Surgical limb: 5% femur, **5 mo 1.6, 10 mo 1.65, 15 mo 1.7, 20 mo 1.75, 25 mo 1.78** 15% femur, 5 mo 1.05, 10 mo 1.04, 15 mo 1.03, 20mo 1.02, 25 mo 1.01 Nonsurgical limb: 5% femur, slight decrease from 1.75 to 1.7 over 5-25 mo 15% femur, constant at 1.1 over 5-25 mo	NR	Reported above	
Jack et al 2023[9]	Intervention (n = 17): rehabilitation with BFR therapy	Mean ± SD, 65 ± 30 d	Distal femur: injured limb BMD 1.19 ± 0.20, BMD healthy limb 1.21 ± 0.19 Proximal tibia: injured limb BMD 1.47 ± 0.16, healthy limb BMD 1.47 ± 0.15 Proximal fibula: injured limb BMD 0.45 ± 0.11, healthy limb BMD 0.47 ± 0.10	LM whole limb: injured limb 8.9 ± 1.5 kg, healthy limb 9.1 ± 1.7 kg LM thigh: injured limb 3.2 ± 0.6 kg, healthy limb 3.3 ± 0.7 kg Bone mass whole limb: injured limb 574.7 ± 101.1 g, healthy limb 588.9 ± 106.6 g Bone mass femur: injured limb 158.2 ± 26.6 g, healthy limb 163.6 ± 28.4 g	6 and 12 wk	Data for injured limb estimated from graph; SD NR. Data for noninjured limb NR. Wk 6: **distal femur 1.16**, proximal tibia 1.44, **proximal fibula 0.42** Wk 12: **distal femur 1.14, proximal tibia 1.44, proximal fibula 0.43**	Distal femur: wk 6, %Δ −2.8% ± 1.3%, Δ g/cm^2 −0.04 ± 0.02; wk 12, %Δ −4.9% ± 1.4%, Δ g/cm^2 −0.06 ± 0.02 Proximal tibia: wk 6, %Δ −1.9% ± 1.7%, Δ g/cm^2 −0.03 ± 0.02; wk 12, %Δ −2.2% ± 1.9%, Δ g/cm^2 −0.04 ± 0.03. Proximal fibula: wk 6, %Δ −7.4% ± 2.9%, Δ g/cm^2 −0.03 ± 0.01; wk 12, %Δ −7.1% ± 3.4%, Δ g/cm^2 −0.08 ± 0.01	**Time to return to sport was significantly shorter in the intervention group (6.42 ± 0.31 mo) compared with the control (8.30 ± 0.54 mo).** Both groups showed similar improvements in all functional measures over 4 wk, except for the anterior Y-balance, where only the **BFR group had a significant improvement between 8 and 12 wk.** Control group experienced **significant decreases in total LM and lower extremity LM at wk 6 and 12, whereas the BFR group maintained LM and showed less bone mass loss.**	After ACLR, BFR may decrease muscle and bone loss for up to 12 wk postoperatively and may improve time to return to sport, with functional outcomes comparable with those of standard rehabilitation.

(continued)

TABLE A1
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		Baseline Testing			Follow-up Testing				
Study	Study Group	Time of Baseline Testing After Injury	BMD, g/cm^2, Mean ± SD	Other Notable Baseline Values, Mean ± SD	Follow-up	BMD, g/cm^2, Mean ± SD	Bone Density Change Score (from baseline)	Other Notable Results (eg, return to activity, function)	Summary
	Control (n = 15): routine rehabilitation	Mean ± SD, 65 ± 36 d	Distal femur: injured limb BMD 1.20 ± 0.22, healthy limb BMD 1.21 ± 0.25 Proximal tibia: injured limb BMD 1.47 ± 0.21, healthy limb BMD 1.46 ± 0.20 Proximal fibula: injured limb BMD 0.47 ± 0.11, healthy limb BMD 0.47 ± 0.12	LM whole limb: injured limb 9.1 ± 1.7 kg, healthy limb 9.3 ± 1.4 kg LM thigh: injured limb 3.2 ± 0.9 kg, healthy limb 3.3 ± 1.0 kg Bone mass whole limb: injured limb 560.6 ± 132.1 g, healthy limb 562 ± 130.9 g Bone mass femur: injured limb 154.3 ± 36.4 g, healthy limb 155.1 ± 37.4 g	6 and 12 wk	Wk 6: **distal femur 1.10, proximal tibia 1.39, proximal fibula 0.4** Wk 12: **distal femur 1.09, proximal tibia 1.36, proximal fibula 0.39**	Distal femur: wk 6, %Δ –7.6% ± 1.4, Δ g/cm^2 −0.10 ± 0.02; wk 12, %Δ –8.2% ± 1.7, Δ g/cm^2 −0.12 ± 0.02 Proximal tibia: wk 6, %Δ –4.8% ± 2.4, Δ g/cm^2 −0.08 ± 0.03; wk 12, %Δ –8.0% ± 1.1, Δ g/cm^2 −0.12 ± 0.03. Proximal fibula: wk 6, %Δ 13.0% ± 3.4, Δ g/cm^2 −0.07 ± 0.01; wk 12, %Δ 15.0% ± 2.5, Δ g/cm^2 −0.08 ± 0.01.	Reported above	
Knurr et al 2022.[12]	Intervention (n = 33): athletes who underwent ACLR	Before injury	5% femur length 1.94 ± 0.03, 15% femur length 1.26 ± 0.03, 50% femur length 2.27 ± 0.04 5% tibia length 1.40 ± 0.03, 15% tibia length 1.40 ± 0.03, 50% tibia length 1.66 ± 0.03	NR	6, 12, 24 mo	Surgical limb: 6 mo post-ACLR: 5% femur length 1.79 ± 0.04, 15% femur length 1.20 ± 0.03, 50% femur length 2.26 ± 0.04, 5% tibia length 1.35 ± 0.03, 15% tibia length 1.31 ± 0.03, 50% tibia length 1.65 ± 0.03 12 mo post-ACLR: 5% femur length 1.87 ± 0.04, 15% femur length 1.21 ± 0.03, 50% femur length 2.31 ± 0.04, 5% tibia length 1.39 ± 0.03, 15% tibia length 1.32 ± 0.03, 50% tibia length 1.68 ± 0.04 24 mo post-ACLR: 5% femur length 1.92 ± 0.04, 15% femur length 1.23 ± 0.03, 50% femur length 2.32 ± 0.05, 5% tibia length 1.40 ± 0.03, 15% tibia length 1.34 ± 0.03, 50% tibia length 1.70 ± 0.04 Nonsurgical limb: 6 mo post-ACLR: 5% femur length 1.92 ± 0.04, 15% femur length 1.24 ± 0.03, 50% femur length 2.24 ± 0.04, 5% tibia length 1.33 ± 0.14, 15% tibia length 1.32 ± 0.04, 50% tibia length 1.67 ± 0.04	Mean ± SE difference from preinjury (% change) Surgical limb: 6 mo post-ACLR: **5% femur length –0.15 ± 0.02 (–7.7%), 15% femur length –0.06 ± 0.01 (–4.8%)**, 50% femur length −0.01 ± 0.02 (–0.4%), 5% tibia length −0.05 ± 0.02 (–3.6%), 15% tibia length −0.09 ± 0.01 (–6.4%), 50% tibia length −0.01 ± 0.01 (–0.6%) 12 mo post-ACLR: 5% femur length −0.07 ± 0.02 (–3.6%), **15% femur length –0.05 ± 0.01 (–4.0%)**, 50% femur length 0.04 ± 0.02 (1.7%), 5% tibia length −0.01 ± 0.02 (–0.7%), **15% tibia length –0.07 ± 0.01 (–5.1%)**, 50% tibia length 0.02 ± 0.02 (1.2%) 24 mo post-ACLR: 5% femur length −0.02 ± 0.02 (–1.0%), **15% femur length –0.03 ± 0.01 (–2.4%),** 50% femur length 0.05 ± 0.02 (2.1%), 5% tibia length 0.00 ± 0.02 (0.0%), 15% tibia length −0.06 ± 0.01 (–4.1%), 50% tibia length 0.04 ± 0.02 (2.4%)	85% of athletes returned to sport by 24 mo.	BMD at the surgical limb's 15% femur region remained 7.1% below preinjury levels at 24 mo post-ACLR. BMD at the 5% femur and 15% tibia regions decreased at 6 and 12 mo but returned to normal by 24 mo, whereas the 50% femur and tibia regions showed no changes.

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TABLE A1
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		Baseline Testing			Follow-up Testing				
Study	Study Group	Time of Baseline Testing After Injury	BMD, g/cm^2, Mean ± SD	Other Notable Baseline Values, Mean ± SD	Follow-up	BMD, g/cm^2, Mean ± SD	Bone Density Change Score (from baseline)	Other Notable Results (eg, return to activity, function)	Summary
						12 mo post-ACLR: 5% femur length 1.89 ± 0.04, 15% femur length 1.25 ± 0.03, 50% femur length 2.30 ± 0.04, 5% tibia length 1.31 ± 0.03, 15% tibia length 1.31 ± 0.04, 50% tibia length 1.68 ± 0.04 24 mo post-ACLR: 5% femur length 1.89 ± 0.04, 15% femur length 1.25 ± 0.03, 50% femur length 2.32 ± 0.05, 5% tibia length 1.31 ± 0.04, 15% tibia length 1.30 ± 0.04, 50% tibia length 1.70 ± 0.04	Nonsurgical limb: 6 mo post-ACLR: 5% femur length −0.02 ± 0.02 (–1.0%), 15% femur length −0.02 ± 0.01 (–1.6%), 50% femur length −0.04 ± 0.02 (–1.8%), 5% tibia length −0.05 ± 0.02 (–3.6%), 15% tibia length 0.01 ± 0.01 (0.8%), 50% tibia length 0.01 ± 0.02 (0.6%) 12 mo post-ACLR: 5% femur length −0.05 ± 0.02 (–2.6%), 15% femur length −0.02 ± 0.01 (–0.6%), 50% femur length 0.00 ± 0.02 (0.0%), 5% tibia length −0.07 ± 0.02 (–5.1%), 15% tibia length 0.00 ± 0.02 (0.0%), 50% tibia length 0.02 ± 0.01 (1.2%) 24 mo post-ACLR: 5% femur length −0.05 ± 0.02 (–2.6%), 15% femur length −0.01 ± 0.01 (–0.8%), 50% femur length 0.04 ± 0.02 (1.8%), 5% tibia length −0.10 ± 0.02 (–7.2%), 15% tibia length −0.01 ± 0.02 (–0.8%), 50% tibia length 0.04 ± 0.02 (1.2%)		
Anderson et al 2019[2]	Intervention (n = 1): rehabilitation program with tailored diet and exercises post-ACL injury	6 d before injury	NR	Injured leg fat mass 2.0 kg Noninjured leg fat mass 2.1 kg Injured leg FFM 11.45 kg Noninjured leg FFM 11.45 kg	6, 12, 18, 28, 38 wk	NR	NR	Injured leg fat mass (kg): 1.8 at 6 wk, 1.7 at 12 wk, 1.9 at 18, 28, and 38 wk Noninjured leg fat mass: 1.8 at 6 and 12 wk, 1.9 at 18, 28, and 38 wk Injured leg FFM (kg): 10.5 at 6 wk, 10.6 at 12 wk, 10.9 at 18 wk, 11.5 at 28 wk, 10.8 at 38 wk Noninjured leg FFM: 10.75 at 6 wk, 10.9 at 12 wk, 11.0 at 18 wk, 11.75 at 28 wk, 11.0 at 38 wk	The athlete experienced a decrease in total body mass and FFM during an initial energy deficit, followed by an increase in body mass with higher carbohydrate intake, suggesting the importance of avoiding excessive energy reduction post-ACL surgery to limit muscle atrophy.

(continued)

TABLE A1
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		Baseline Testing			Follow-up Testing				
Study	Study Group	Time of Baseline Testing After Injury	BMD, g/cm^2, Mean ± SD	Other Notable Baseline Values, Mean ± SD	Follow-up	BMD, g/cm^2, Mean ± SD	Bone Density Change Score (from baseline)	Other Notable Results (eg, return to activity, function)	Summary
Kroker et al 2019[14]	Intervention (n = 15): young adults with acute unilateral ACL tears	Mean 29 d after injury (range, 7-43 d)	Estimated from graph; SD NR. Injured femur aBMD 1.2 Injured tibia aBMD 1.23 Contralateral femur 1.2 Contralateral tibia 1.2	NR	100, 200, 300 d	Injured femur aBMD 100 d, 1.15 Injured tibia aBMD 100 d, 1.15 Injured femur aBMD 200 d, 1.1 Injured tibia aBMD 200 d, 1.1 Injured femur aBMD 300 d, 1.15 Injured tibia aBMD 300 d, 1.14 Contralateral femur at 100-300 d, 1.2 Contralateral tibia at 100-300 d, 1.2	NR	Bone loss in the femur and tibia peaked at 206 and 218 d, respectively. Traumatic BMLs were seen in 13 of 15 participants, resolving 90% by 4 mo and fully by 8 mo. Bone loss was more severe in BML areas, with reduced trabecular number and thickness and increased separation.	Significant bone loss occurred in the injured knee after acute ACL tears, peaking at 215-254 d after injury, followed by a plateau or recovery. The lateral femoral compartment showed subchondral bone plate thinning, especially at BML sites, suggesting that early intervention may prevent posttraumatic osteoarthritis.
Milsom et al 2014[19]	Intervention (n = 1): 8-wk immobilization and physical therapy, followed by diet-controlled, weightbearing rehabilitation program	3 d	NR	Immobilized leg LM 10.8 kg Immobilized leg fat mass 2.0 kg Nonimmobilized leg LM 10.8 kg Nonimmobilized leg fat mass 1.8 kg	8, 12, 16, 20, 24 wk	NR	NR	Immobilized leg LM (kg): 9.4 at 8 wk, 9.3 at 12 wk, 10.2 at 16 wk, 10.5 at 20 wk, 10.6 at 24 wk Immobilized leg fat mass (kg): 2.3 at 8 wk, 2.2 at 12 wk, 1.8 at 16 wk, 1.9 at 20 wk, 1.85 at 24 wk Nonimmobilized leg LM: 9.9 at 8 wk, 9.5 at 12 wk, 11 at 16 wk, 10.85 at 20 wk, 10.9 at 24 wk Nonimmobilized leg fat mass: 2.2 at 8 wk, 1.8 at 12 wk, 2.0 at 16 wk, 1.8 at 20 wk, 1.7 at 24 wk	During 8 wk of inactivity and immobilization on a low-carbohydrate–high-protein diet, the athlete lost 5 kg total mass, including 5.8 kg LM, with a 13% LM loss in the immobilized limb (1.4 kg). From wk 8 to 24, with a moderate-carbohydrate–high-protein diet and structured training, the athlete gained LM, with the immobilized limb showing gradual gains (1.2 kg), improving their overall physical profile.

(continued)

TABLE A1
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		Baseline Testing			Follow-up Testing				
Study	Study Group	Time of Baseline Testing After Injury	BMD, g/cm^2, Mean ± SD	Other Notable Baseline Values, Mean ± SD	Follow-up	BMD, g/cm^2, Mean ± SD	Bone Density Change Score (from baseline)	Other Notable Results (eg, return to activity, function)	Summary
Zerahn et al 2006[32]	Intervention (n = 18): 2-mo physical therapy, ACLR, structured 3-mo rehabilitation with physical therapy	69 mo (range, 4-162 mo)	Raw BMD values NR for intervention group at baseline	NR	4, 12, 24 mo	NR	Raw values depicted in graph and could not be estimated. **Significant declines in *Z* scores were observed in the proximal tibia of the operated leg at 4 and 12 mo compared with controls and the noninjured leg.** **The lateral tibia also had significantly lower *Z* scores at 24 mo.** The medial tibia's *Z* score did not significantly differ from controls at 24 mo, although there was a trend toward lower scores on the injured side. No significant changes were found in the calcaneus and noninjured leg.	Knee performance on sport score increased from a median of 2 (abnormal) to 1 (almost normal). Knee performance on activity of daily living increased from a median of 2 (abnormal) to 0 (normal). At the 24-mo follow-up, patients reported improved knee performance in sports activities, associated with increases in *Z* scores for both the medial **(r = 0.59)** and lateral tibia **(r = 0.54)**.	There was a partially reversible decline in BMD of the proximal tibia after arthroscopic ACLR. Improvement in knee performance in sports activities was associated with an increase in BMD of the injured leg.
	Control: healthy adults	Reported above	Medial tibia 1.060 ± 0.144, lateral tibia 0.913 ± 0.120, calcaneus 0.674 ± 0.101	NR	4, 12, 24 mo	NR	Reported above	NR	

[a]Bolded values represent significant changes as reported by studies. aBMD, areal bone mineral density; ACL, anterior cruciate ligament; ACLR, anterior cruciate ligament reconstruction; BFR, blood flow restriction; BMD, bone mineral density; BML, bone marrow lesion; DEXA, dual-energy x-ray absorptiometry; FFM, fat-free mass; LM, lean mass; NR, not reported; PKEM, peak knee extensor movement; PKF, peak knee flexion; PT, peak torque; ROI, region of interest; RTD, rate of torque development;

TABLE A2
Summary of Outcomes Examined by Included Studies on Non-ACL Injuries[a]

		Baseline Testing			Follow-up Testing				
Study	Study Group	Time of Baseline Testing After Injury	BMD, g/cm^2, Mean ± SD	Other Notable Baseline Values, Mean ± SD	Follow-up	BMD, g/cm^2, Mean ± SD	Bone Density Change Score (from baseline)	Other Notable Results (eg, return to activity, function)	Summary
Alway et al 2022[1]	Intervention (n = 17): injured players receiving LSF rehabilitation	11 underwent DEXA scans on date of diagnosis, 2 scanned within 14 d of diagnosis, and 4 scanned before diagnosis	L1-L4 1.464 ± 0.150, L3 1.634 ± 0.170, CL3 1.795 ± 0.215, IL4 1.633 ± 0.184, **CL4 1.852 ± 0.195**	Fat-free mass was 69.01 ± 3.85 kg, and career match workload was 3887 ± 4217 balls.	Short term (5-11 wk after baseline or diagnosis), medium term (15-34 wk) and long term (52-59 wk)	Depicted in graph. Raw values could not be estimated.	BMD decreased for up to 21-24 wk after diagnosis (L1-L4, 2.3%; IL3, 1.9%; CL3, 3.0%; IL4, 2.2%; CL4, 2.0%) and increased to above baseline levels by 52 wk (L1-L4, 2.2%; IL3, 0.7%; CL3, 1.8%; IL4, 1.9%; CL4, 2.1% greater than baseline).	Fast bowlers with LSF demonstrated **significantly lower L1-L4 *Z* scores (−0.99) and L1-L4 BMD at diagnosis (7.6%)**, which was accompanied by deficits on the contralateral side of L3 and L4 (8.8% and 10.4%, respectively).	Fast bowlers with LSF had lower lumbar BMD than uninjured bowlers, particularly on the contralateral side, decreasing up to 3.0% by 20-24 wk and recovering by 52 wk.
	Control (n = 12): uninjured players	Reported above	L1-L4 1.585 ± 0.134, L3 1.703 ± 0.175, CL3 1.968 ± 0.256, IL4 1.711 ± 0.157, and **CL4 2.067 ± 0.335**	Fat-free mass was 71.13 ± 7.40 kg, and career match workload was 3933 ± 3767 balls.	Short term (5-11 wk after baseline or diagnosis), medium term (15-34 wk) and long term (52-59 wk)	Depicted in graph. Raw values could not be estimated.	Small increases in BMD across time at L1-L4, IL3, and CL3 sites, with 1.6%, 1.8%, and 2.0% increases at 52 wk, respectively, compared with baseline. IL4 and CL4 increases observed for the first 28-30 wk (2.1% and 0.7% respectively), before decreasing to near baseline levels at 52 wk (0.5% and 0.3%, respectively).		
Kiebzak 2018[11]	Intervention (n = 1): diet program, exercise on a treadmill and stationary bike	12 mo	BMD 1.245	Fat mass: 29.02 kg LM 39.61 kg	12 mo	BMD 1.230	NR	Fat mass 22.58 kg LM 41.98 kg	
Mueller et al 2015[20]	Intervention (n = 1): hip arthroscopy, followed by immobilization and gradual increase in training	4 y before injury (T0) and 1 wk after surgery (T1)	NR	NR	8 wk after surgery (T2), 14 wk after surgery (T3), 9 mo after surgery (T4)	NR	Total aBMD in the left proximal femur **increased by 6.2% from T0 DEXA to T1, then decreased by 2.0% from T1 to T2. It rose by 1.7% from T2 to T3 and by 1.0% from T3 to T4.** In the right proximal femur, aBMD **increased by 5.4% from T0 DEXA to T1, with a 3.6% increase in the trochanter and a 2.7% decrease in the shaft from T1 to T2.** In the lumbar spine, total aBMD **increased by 5.6% from T0 DEXA to T1, decreased by 1.6% from T1 to T2, and returned to presurgery values by T4, with changes mainly in L1 and L3.**	Left lower extremity LM increased from T0 DEXA to T1, decreased by 4.5% during immobilization, then rose by 4.5% from T2 to T3 and by 6.9% from T3 to T4. Right lower extremity LM increased 13.1% from T0 DEXA to T4, with a pause between T1 and T2. Left leg LM dropped 12.7% from T1 to T2, then increased 9.9% from T2 to T3 and 13.4% from T3 to T4. Right leg mass was constant until T2, then increased 8.1% from T2 to T3 and 10.6% from T3 to T4. Most mass loss from T1 to T2 was in the legs.	The DEXA assessment showed a 22% fat reduction and 6% LM increase, reflecting successful diet and exercise. After hip arthroscopy, aBMD and LM losses in the lower extremity and lumbar spine were fully reversed after 6 wk of training, with a professional Ironman recovering presurgery levels in 4 mo and maintaining them at 9 mo.

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TABLE A2
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		Baseline Testing			Follow-up Testing				
Study	Study Group	Time of Baseline Testing After Injury	BMD, g/cm^2, Mean ± SD	Other Notable Baseline Values, Mean ± SD	Follow-up	BMD, g/cm^2, Mean ± SD	Bone Density Change Score (from baseline)	Other Notable Results (eg, return to activity, function)	Summary
Magnusson et al 2003[17]	Intervention (n = 14): athletes with MTSS	29 mo (range, 5-120 mo)	Total body, 1.26 ± 0.05; femoral neck, 1.17 ± 0.10; Ward triangle, 1.13 ± 0.15; trochanter region, 1.06 ± 0.11; tibia region 1, 1.27 ± 0.13; tibia region 2, 1.52 ± 0.15; tibia region 3, 1.92 ± 0.23; tibia region 4, 1.39 ± 0.19; tibia region 5, 1.33 ± 0.17	NR	Mean ± SD 5.7 ± 1.3 y	Total body, 1.30 ± 0.07; femoral neck, 1.13 ± 0.14; Ward triangle, 1.06 ± 0.20; trochanter, 1.03 ± 0.13; tibia region 1, 1.28 ± 0.14; tibia region 2, 1.56 ± 0.14; tibia region 3, 1.93 ± 0.20; tibia region 4, 1.65 ± 0.22; tibia region 5, 1.30 ± 0.15	Change per year: **total body, 0.008 ± 0.005; femoral neck, −0.007 ± 0.018; Ward triangle, −0.014 ± 0.014;** trochanter, −0.004 ± 0.008; tibia region 1, 0.002 ± 0.012; **tibia region 2, 0.007 ± 0.019;** tibia region 3, 0.000 ± 0.019; **tibia region 4, 0.046 ± 0.023;** tibia region 5, −0.004 ± 0.020	During the follow-up, athletes with MTSS experienced a significant increase in BMD by **19% ± 11%** in the region corresponding to their pain. The annual increase in BMD in the painful region for athletes with MTSS was **3.4%, significantly higher than the −0.1%** change observed in nonathlete control subjects.	The localized low BMD in the tibial region corresponding to the pain increased after recovery in athletes with MTSS.
	Control (n = 24): nonathletes (without MTSS)	Reported above	Total body, 1.23 ± 0.07; femoral neck, 1.08 ± 0.17; Ward triangle, 1.04 ± 0.20; trochanter region, 0.96 ± 0.14; tibia region 1, 1.16 ± 0.16; tibia region 2, 1.52 ± 0.18; tibia region 3, 1.87 ± 0.18; tibia region 4, 1.57 ± 0.15; tibia region 5, 1.22 ± 0.15	NR	Mean ± SD 4.0 ± 0.7 y	Total body, 1.24 ± 0.07; femoral neck, 1.04 ± 0.15; Ward triangle, 0.99 ± 0.18; trochanter, 0.95 ± 0.15; tibia region 1, 1.19 ± 0.16; tibia region 2, 1.55 ± 0.15; tibia region 3, 1.99 ± 0.16; tibia region 4, 1.59 ± 0.13; tibia region 5, 1.23 ± 0.13	Change per year: **total body, −0.003 ± 0.008; femoral neck, −0.025 ± 0.018; Ward triangle, −0.026 ± 0.018;** trochanter, −0.012 ± 0.018; tibia region 1, −0.011 ± 0.022; **tibia region 2, −0.013 ± 0.028;** tibia region 3, −0.004 ± 0.041; **tibia region 4, −0.004 ± 0.038;** tibia region 5, −0.009 ± 0.026	Reported above	

(continued)

TABLE A2
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		Baseline Testing			Follow-up Testing				
Study	Study Group	Time of Baseline Testing After Injury	BMD, g/cm^2, Mean ± SD	Other Notable Baseline Values, Mean ± SD	Follow-up	BMD, g/cm^2, Mean ± SD	Bone Density Change Score (from baseline)	Other Notable Results (eg, return to activity, function)	Summary
Therbo et al 2003[28]	Intervention (n = 12): patients with Achilles rupture undergoing surgical repair and immobilization	7 d after surgery	BMC for proximal tibia, BMD for femoral neck and greater trochanter. Values are mean (range). Injured leg: Proximal tibia 8.572 (6.635-10.55) Femoral neck 1.104 (0.788-1.364) Greater trochanter 0.972 (0.731-1.246) Healthy leg: Proximal tibia 8.664 (6.446-10.060) Femoral neck 1.086 (0.798-1.328) Greater trochanter 0.971 (0.669-1.176)	NR	6 wk, 3 mo, 6 mo, 1 y	Injured leg: 6 wk, **proximal tibia 8.431 (5.982-10.920), femoral neck 1.091 (0.754-1.413), greater trochanter 0.923 (0.613-1.180)** 3 mo, **proximal tibia 8.219 (5.507-10.380), femoral neck 1.069 (0.745-1.361), greater trochanter 0.907 (0.630-1.160)** 6 mo, **proximal tibia 8.153 (5.852-10.600), femoral neck: 1.073 (0.772-1.344), greater trochanter: 0.893 (0.642-1.094)** 1 y, **proximal tibia 8.049 (5.952-10.600), femoral neck 1.072 (0.771-1.315), greater trochanter 0.904 (0.639-1.087)** Healthy leg: 6 wk, **proximal tibia 8.765 (6.230-10.930),** femoral neck 1.094 (0.815-1.344), greater trochanter 0.981 (0.671-1.336) 3 mo, **proximal tibia 8.429 (6.308-10.030),** femoral neck 1.066 (0.814-1.308), greater trochanter 0.918 (0.655-1.180) 6 mo, **proximal tibia 8.206 (6.319-9.725),** femoral neck 1.062 (0.790-1.322), greater trochanter 0.960 (0.654-1.192) 1 y, **proximal tibia 8.263 (5.972-10.280),** femoral neck 1.062 (0.799-1.312), greater trochanter 0.951 (0.675-1.166)	Mean decrease in bone mineral 1 y after the injury was 6.4%, 2.5%, and 6.8% for measurements in, respectively, the proximal tibia, femoral neck, and greater trochanter.	NR	One year after an Achilles tendon rupture, patients showed a significant decrease in BMC of the proximal tibia and a decrease BMD at the femoral neck and greater trochanter. These findings suggest that such patients may be at an increased risk for osteoporotic fractures.

[a]ACL, anterior cruciate ligament; BMC, bone mineral content; BMD, bone mineral density; CL, contralateral lumbar; DEXA, dual-energy x-ray absorptiometry; IL, ipsilateral lumbar; LM, lean mass; LSF, lumbar stress fracture; MTSS, medial tibial stress syndrome; NR, not reported; ROI, region of interest.